साक्षी

Sākṣī

The witness: one who sees a thing directly, and can testify to it.

HACKATHON THEME

What if Buddha were born in 2026? Build the future guided by the timeless wisdom of Lord Gautam Buddha.

TEAM

Aaditya Sapkota / AI Backend Engineer
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SUBMISSION

Repository
github.com/LumbiniX-Committee/Everest

Live build
application-a925ec89-66ca-48ae-a9c8-7440674a46082.apk

Demo video
drive.google.com/drive/folders/1by7PGKVBpTmufWhx-YoBqBDkdZ02BofS

DOCUMENT STATUS

Every figure in this document was verified against the working tree on 10 August 2026. Where a capability is implemented but not yet demonstrated on the presentation device, this document says so rather than implying otherwise. Section 7 and Section 5.6 record the open gaps in full.

Contents

THE ONE IDEA, IF YOU READ NOTHING ELSE

The Buddha's first instruction to the Kālāmas was not to accept a teaching because it is traditional, because it is repeated, or because a teacher said it. Test it, and see for yourself. Sākṣī is that instruction rendered as software: an app that never claims more than it can support. A photograph matched by eye is not recorded as a sensor lock. A missing sensor reading is stored as unknown, never as zero. An answer that cannot cite a canonical passage is a refusal, not a paragraph. A detected crack is a candidate until a person confirms it. A documentary coordinate is never presented as a survey.

Sākṣī (Sanskrit: *sākṣin*), one who is present at an event and can bear witness to it.

Executive Summary

Project Vision

In 2026, the technology that surrounds a visitor to Lumbini is optimised to hold attention. The teaching that was first given at Lumbini is about placing attention deliberately. Sākṣī is built on the belief that these need not be opposed: that a phone can be an instrument of attention rather than a competitor for it, and that the most modern thing we can do with machine learning near a sacred site is to make it admit what it does not know.

If the Buddha were born in 2026, the sermon would not need a new medium so much as a new discipline. The Kālāma Sutta asks a listener not to accept a claim on the strength of tradition, repetition, rumour, scripture, logic, inference, appearance, agreement with one's own views, apparent competence, or the authority of a teacher, but to test it. That is a specification, and it is an unusually demanding one for software built in the age of the confident language model. We took it literally. Every subsystem in this application is designed so that a claim it makes can be traced to something a person can check: a canonical passage id, a timestamped photograph, a sensor reading with its accuracy attached, or a coordinate with its provenance tier printed beside it.

The name states the thesis. *Sākṣī* means witness: someone who sees a thing directly and can testify to it. A witness who embellishes is worse than no witness at all.

The Pitch

Sākṣī is a map-based Android application for the Greater Lumbini Circuit that turns a visitor's attention into conservation evidence. A pilgrim or tourist opens the map, walks to one of twelve heritage sites, and receives that place's own material at a depth that grows as they return. At each site there are fixed photographic viewpoints: the app guides the visitor to stand where an earlier observer stood, aligns the device against a stored heading and tilt, and records what is there today. Those repeated frames become a comparable time-series that a conservation authority can actually use, and any damage a visitor notices becomes a structured condition report with a photograph, a location and a severity attached. Alongside it sits a question-answering surface restricted to a narrow canonical Buddhist corpus, which answers in Nepali or English with a citation to a specific passage, and refuses when the sources do not support an answer. The app is built for three audiences at once: international pilgrims and local travellers who want context that is true, and the Lumbini Development Trust and municipal heritage officers who currently have no continuous record of how these monuments are changing.

3	12	113/113	49/50	100%
PRODUCT SURFACES	HERITAGE SITES SEEDED	DOMAIN TESTS PASSING	DHAMMA BENCHMARK	ADVERSARIAL REFUSAL

The application is offline-first by design, because the Sacred Garden has patchy signal and because a photograph taken at a vantage on a particular day cannot be retaken if an upload fails. Every observation is written to the device database first and synchronised opportunistically afterwards. On-device machine learning does the same work without a network: a crack detector runs entirely on the phone, and a small quantised language model can answer with the radio switched off.

THE THREE SURFACES AT A GLANCE (EXPANDED IN SECTION 4)

SURFACE	THE QUESTION IT ANSWERS	WHAT IT PRODUCES
Tīrtha the journey	What am I looking at, and what happened here?	Arrival-aware context that deepens across four wisdom tiers, historical comparison imagery, and a restrained quest layer.
Sākṣī the witness	What does this look like today, compared with last time?	Aligned, dated, positioned photographs at fixed vantages, plus structured condition reports with severity and corroboration.
Dhamma the truth	What do the sources actually say, and where?	Answers in Nepali or English carrying a canonical passage identifier, or an explicit refusal naming the missing evidence.

Lumbini receives well over a million visitors a year across a site complex spanning several square kilometres, with monuments dating from the third century BCE.

Problem Statement

2.1 The Disconnect

A visitor standing in front of the Māyādevī Temple has three options for understanding what they are looking at. They can read a weathered interpretive board written for a general audience a decade ago. They can hire a guide, whose quality is uneven and whose availability is seasonal. Or they can search the internet, which returns a mixture of tourism marketing, devotional writing, and confident summaries of uncertain provenance. None of these is anchored to where the person is standing, none of them adapts to whether this is their first visit or their fifth, and none of them can tell them which claims are archaeologically established and which are traditional.

The result is a specific kind of loss. A person travels a long way to a place of genuine significance and leaves with a photograph and a vague impression. The Ashokan pillar, which carries the inscription that identifies this site as the birthplace, becomes a stone column in a queue. The nativity sculpture becomes a thing behind glass. The material is extraordinary and the encounter is thin, because nothing bridges the two at the moment the person is actually there.

2.2 Heritage Maintenance

The conservation problem is quieter and more serious. The monuments of the Greater Lumbini Circuit are spread across a wide area, several of them are excavated brick structures exposed to monsoon, and the authorities responsible for them have a small staff relative to the ground they cover. Condition assessment is therefore episodic: a specialist survey happens, produces a report, and the next one happens years later. Between those points there is no continuous record.

This matters because most heritage degradation is not dramatic. It is a crack widening by millimetres a season, biological growth spreading across a brick face, water pooling where drainage has silted up, and salt efflorescence appearing after a wet year. These are visible to anyone who looks, and invisible to anyone who is not comparing against a previous image from the same position. Meanwhile several hundred thousand people a year photograph these monuments from roughly the same spots, and none of that imagery is comparable, positioned, or retained.

> **THE SPECIFIC GAP**
>
> There is no cheap, continuous, positionally comparable photographic record of the Lumbini monuments, despite an enormous volume of photography happening at them every day. The imagery exists. What is missing is the discipline of standing in the same place, the metadata to prove you did, and somewhere for it to go.

2.3 Misinformation

The third problem arrived recently and is getting worse. General-purpose language models answer questions about Buddhist doctrine and Lumbini's history fluently, at length, and with no reliable signal of when they are reconstructing from pattern rather than reporting from a source. The failure modes are specific and damaging in this domain: they invent sutta references and attribute quotations to passages that do not contain them; they blend traditions, presenting a later Mahāyāna or Tibetan formulation as if it were an early Pāli teaching; they will adopt the voice of the Buddha and generate first-person scripture on request; and they are markedly worse in Nepali than in English, which means the people closest to the site get the least reliable answers.

For a religious and historical corpus this is not a quality-of-service issue. A fabricated citation that looks like a real one is worse than a refusal, because it is repeatable, quotable, and indistinguishable from a genuine reference to a reader who does not have the canon open. Every safeguard in our knowledge surface is a response to a failure mode we reproduced first.

The app has exactly three destinations. There is no Home, Explore, Rewards or Profile tab. Adding a fourth would require an argument.

Proposed Solution

3.1 Overview

Sākṣī is a map-based Android application organised around three surfaces, each answering one question a person actually has while standing in Lumbini. The three surfaces are the conceptual model of the product, not a navigation convenience.

Tīrtha THE JOURNEY · THE PLACE

A live map of the Greater Lumbini Circuit with real-time position, arrival detection at twelve seeded sites, site detail, historical Then and Now plates, narration, and a restrained quest layer. It answers: *what am I looking at, and what happened here?*

Sākṣī THE WITNESS · THE EVIDENCE

Fixed-point rephotography. Walk to a vantage, align heading and tilt against the stored target, capture, and file a structured condition report. A register holds the resulting time-series, and a global leaderboard ranks contributors. It answers: *what does this look like today, compared with the last time somebody stood here?*

Dhamma THE TRUTH · THE SOURCES

Citation-locked question answering over a deliberately narrow canonical corpus, in Nepali by default with an English toggle. It refuses when the sources are thin, and it will not impersonate the Buddha. It answers: *what do the texts actually say, and where exactly do they say it?*

3.2 Value Proposition

The bridge between past and future in this application is a single mechanism used three ways: **an attentive person, standing in a specific place, produces something durable.**

WHAT EACH AUDIENCE GIVES, AND WHAT EACH RECEIVES

AUDIENCE	WHAT THEY GET	WHAT THEY CONTRIBUTE WITHOUT EXTRA EFFORT
Pilgrims international and devotional	Context anchored to where they stand, deepening with return visits; answers about doctrine that carry a verifiable citation; a practice register rather than a score.	A dated, positioned photograph of a monument, and any condition they noticed while looking closely.
Tourists domestic and regional	A guided route, historical comparison imagery that makes the ruins legible, and a reason to look carefully rather than to photograph and move on.	The same photographic record, plus corroboration of reports other visitors filed.
Heritage authorities LDT, municipalities, DoA	A continuous, positionally comparable condition record they did not have to staff, with structured reports triaged by severity and corroboration count.	Survey-grade coordinates and vantage definitions, which is the one input only they can provide.

The reciprocity matters to the design. We deliberately did not build a system where the visitor is unpaid labour dressed as a game. The photographic contribution is a by-product of an experience that is worth having on its own terms, and the merit system that acknowledges it is explicitly non-transferable, non-purchasable and capped, so that it can never become the reason someone visits.

> THE TIMELESS WISDOM, MADE OPERATIONAL
>
> **Right view** becomes a citation validator: no claim reaches the reader without a source. **Right speech** becomes a vocabulary linter that fails the build if engagement or gambling language reaches a screen. **Non-attachment** becomes a merit ledger with no transfer function, no expiry and no purchase path. **Mindfulness** becomes a stillness practice that requires the screen to be off. These are not metaphors bolted onto a feature list; each one is enforced in code and each one is tested.

Seeded content: 12 sites, 6 vantages, 10 quests, 10 historical plates, 5 timeline entries, 3 conservation needs.

Core Modules & MVP Features

4.1 Tīrtha — The Journey

Tīrtha is the place layer. It opens on a vector map of the Greater Lumbini Circuit with the visitor's live position, twelve seeded heritage sites, and their precinct groupings. Sites carry real attributes rather than decorative ones: an evidence tier, a zone classification, a photography policy (some areas of the Sacred Garden restrict or prohibit photography, and the app enforces this rather than relying on signage), and a geofence radius that ranges from thirty metres for the Ashokan pillar to forty-five metres for the temple.

REAL-TIME, LOCATION-BASED MAPPING

- **Arrival detection with hysteresis.** The geofence watcher fires an `enter` event at radius r and an `exit` only at $r \times 1.15$, so a visitor standing on a boundary does not trigger a storm of notifications. A separate `dwell` event fires after sixty continuous seconds inside, which is what unlocks the deeper material.
- **Nearest-site guidance.** A persistent card names the nearest site and the distance to it, so the map is useful while walking rather than only while stopped.
- **Pradakṣiṇā tracking.** Clockwise circumambulation of a stupa is scored from the walked track: completion requires at least 330 degrees clockwise with no more than 30 degrees of reverse travel. Walking anticlockwise is never treated as a failure. It returns a gentle directional hint, because the practice is the point and a scolding notification is not.
- **Silence on consecrated ground.** Inside a designated zone the app suppresses its own notifications. A device that buzzes during someone else's prostration is a design failure regardless of what the notification says.

THE TIERED WISDOM SYSTEM

How much a place says depends on how much attention the visitor has given it. This is a deliberate inversion of the usual pattern, where the app front-loads everything and hopes something is read.

TIER	UNLOCKED BY	WHAT THE PLACE SAYS
Basic	Arrival inside the geofence	Identification, date range, and the one fact that makes the place legible: what it is and why it is here.
Medium	Dwell, or a completed observation	The site's own story beats: excavation history, what the evidence tier actually rests on, the Then and Now comparison.
High	Repeat visits, quest completion, contributed observations	Epigraphic detail, contested interpretations stated as contested, source references, and the conservation status of the fabric.
Custom	Asking	An on-site guide answers free-form questions about the building in front of the visitor, in Nepali or English, at the depth they ask for.

The guide deliberately does not run through the Dhamma engine. We tried that first, and it was instructive: a visitor standing in front of a monastery asking "what is this building" received the answer *"I do not have enough reliable evidence to answer this confidently,"* followed by a reading list. A true sentence, and a useless one. Grounding rules calibrated for scripture are wrong for architecture, so the guide is a separate voice with its own limits, sharing no code path with the citation engine.

THE QUEST LAYER

Quests are restrained on purpose: ten of them, categorised as survey, epigraphy, ecology or monastic, each with a difficulty, an estimated duration, an optional prerequisite, and a availability window evaluated against proximity and time of day. Three mechanics are worth naming. **Observation riddles** give a hint on a wrong answer and never a penalty. **Stillness** requires the screen off, low accelerometer variance, and presence inside a geofence, held for a target duration: the phone-down quest, which the app cannot fake and cannot rush. **The close ritual** offers a card after twenty minutes inside the Sacred Garden suggesting the session end, which is the only feature in the application designed to reduce its own usage.

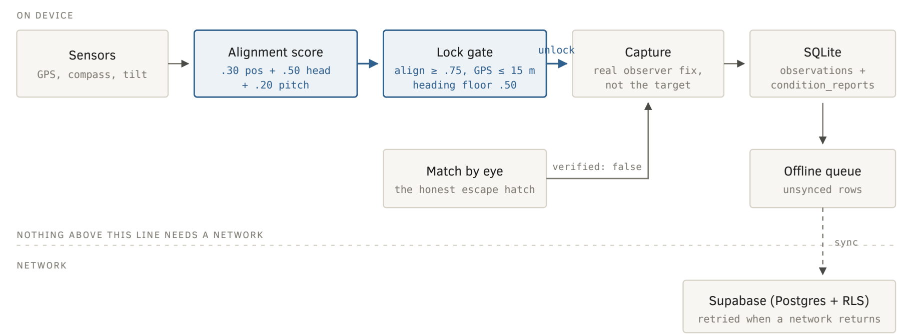
Alignment weighting: 0.30 position, 0.50 heading, 0.20 pitch. Lock needs align ≥ 0.75, GPS accuracy ≤ 15 m, and heading at least half satisfied.

4.2 Sākṣī — The Witness

This is the module the project is named after and the reason the rest of it exists. Fixed-point rephotography is an established conservation technique: return to a defined viewpoint, reproduce the framing, and the difference between two images becomes measurable rather than impressionistic. It is normally expensive because it requires a trained surveyor to travel. Sākṣī makes it something an ordinary visitor can do correctly, by moving the discipline into the device.

BEFORE-AND-AFTER MONUMENT COMPARISON

Two comparison modes are implemented. **Then and Now** pairs a historical plate (archival photographs and documented reconstructions, each carrying its evidence tier and licence provenance) against modern imagery, presented through a scrubber. Where an honest alignment between an archival plate and a modern frame is not possible, the app shows the pair unaligned and says so rather than warping one to fit the other. **The observation series** is the live product: every capture at a given vantage, in date order, from the same position, so that change over a season is visible by scrubbing rather than by argument.



The alignment score is computed once and governs both the on-screen reticle and whether the shutter unlocks, so there is only ever one truth about how well aligned a capture is. Heading carries the heaviest weight because a perfect GPS fix pointed at the wrong wall is not an aligned photograph, while GPS is the noisiest axis and must not be able to veto an otherwise good alignment.

THE REPORTING MECHANISM FOR HERITAGE ISSUES

A condition report is structured, not free text. It carries a category (structural, biological, water, surface, vegetation), a subtype, a severity, an optional note, the photograph, the observer's real position, and a link to the observation it came from. Reports are written to the device database immediately and enter the same offline queue as the captures.

An on-device damage detector assists this. A YOLOv8 crack detector, exported to ONNX and running through the phone's neural runtime, proposes candidate regions on the captured photograph. It offers them to the surveyor and it never writes a report on its own. When the model file or the native runtime is absent, the surface says which of the two is missing and offers the manual path, because a feature that stays silent when it cannot run is indistinguishable from one that was never built. Corroboration is a first-class action: a second visitor confirming an existing report is worth recording, and is separately rewarded.

MERIT, AND THE GAMIFIED GLOBAL LEADERBOARD

Two scoring systems exist and they are deliberately kept apart, because conflating them would undo the property the schema exists to preserve.

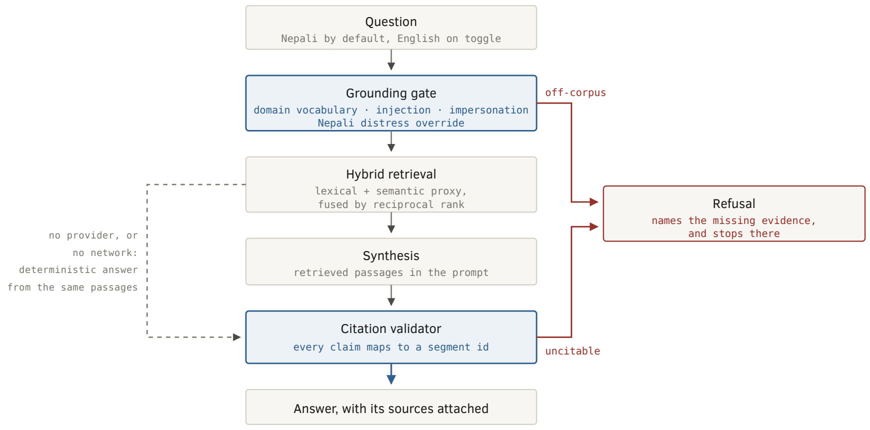
	PUṆYA (MERIT)	GLOBAL LEADERBOARD
Lives	On the device, unsynced	In Postgres, global
Ranked	Never. There is no total shown as a score.	Yes, openly competitive
Computed	Derived from an append-only event log	Server-side, from evidence that already synced
Transferable	No transfer function exists in the codebase	Not applicable
Anti-gaming	200 per day cap; one merit-earning resurvey per vantage per 24 hours; severity never scales a reward	There is no `submitScore` endpoint. Points cannot be claimed, only earned by uploading work.

The merit values are fixed and public: a resurvey pays 50 regardless of what it finds, because "nothing has changed" is a valuable observation and paying more for damage would pay people to find damage. Corroboration and a first report each pay 25. An attention quest pays 70, a path quest 40, a contribution such as a translation or a transcription 30.

Canonical segment ids look like `dn16:6.7` or `sn56.11:4.2`. A reader can open the passage and check.

4.3 Dhamma — The Truth

Dhamma is a question-answering surface restricted to a deliberately narrow canonical corpus, built on segment-aligned Bilara source texts. The narrowness is the feature. A wide corpus invites the system to answer questions it should decline, and every additional tradition admitted multiplies the chance of a subtle attribution error that no reader will catch.



The bypass is the important edge. A missing provider degrades the prose; it never removes the citation. Every path out of this pipeline is either cited or a refusal, which is why adversarial refusal can be reported as 100 percent rather than as "usually".

THE FOUR SAFEGUARDS

1. **The grounding gate, before anything is generated.** A domain vocabulary check establishes whether the question is even in scope. A prompt-injection and impersonation check catches "ignore your instructions" and "answer as the Buddha in the first person". A Nepali distress and self-harm detector bypasses the entire pipeline and returns helpline information, because that is not a retrieval problem and must not be handled as one.
2. **Retrieval before generation, always.** The model never answers from its own parameters. It is given retrieved passages and asked to write from them. Retrieval fuses a lexical ranking with a semantic ranking using reciprocal rank fusion, which gives usable recall with no index server and no embedding cost, and therefore runs identically inside the app and inside the backend.
3. **The citation validator, after generation.** Every claim in the drafted answer must map to a segment id that was actually retrieved. An answer that cannot be validated does not get softened or hedged. It becomes a refusal that names what is missing.
4. **A deterministic cited fallback.** If no model provider is reachable, the engine assembles an answer directly from the retrieved passages. The prose is plainer. The citations are identical. A missing API key degrades an answer; it never fabricates one.

STRICTLY ROOTED IN THE FOUR NOBLE TRUTHS

The corpus is anchored on the Dhammacakkappavattana Sutta (SN 56.11), which sets out the four truths and the eightfold path, and the Mahāparinibbāna Sutta (DN 16), together with the Kālāma Sutta's instruction on testing claims and a small set of related discourses. Questions that sit outside this material are refused with a statement of what the corpus does and does not contain, rather than answered from a wider tradition without saying so. Reflection is offered as a companion mode, framed as inquiry rather than advice: it asks questions back rather than telling a person what to do, because the corpus does not license personal counsel and the honest position is to say so.

BENCHMARK: 50 QUESTIONS, ALL MANDATORY GATES PASSING

CATEGORY	SCORE	WHAT IT TESTS
Answerable	18 / 18	Questions the corpus genuinely supports
Adjacent	9 / 10	Near the corpus edge, where over-answering is the risk
Out of scope	12 / 12	Must refuse, and must say why
Adversarial	6 / 6	Injection and impersonation attempts
Nepali	4 / 4	Nepali intent handling, including the distress override
Overall	**49 / 50**	Refusal precision and recall, and citation hit rate, all above threshold

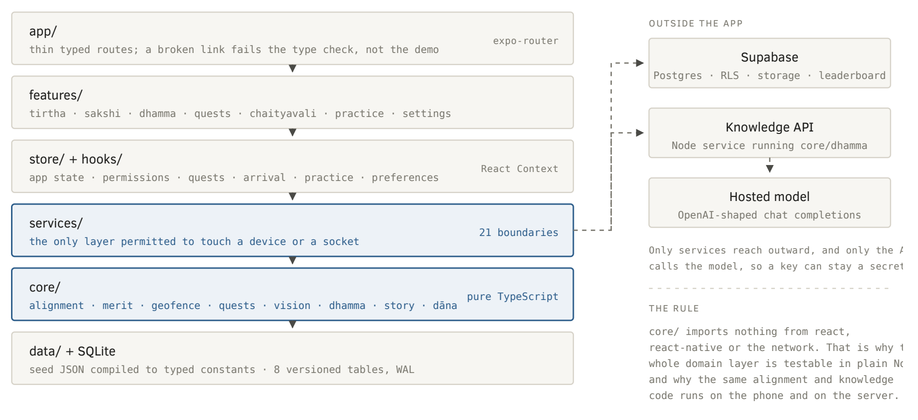
The layering rule: `core/` imports nothing from React, React Native, or the network. It computes; the app decides what to render and when to save.

Technical Architecture & Stack

A NOTE ON THE STACK, STATED PLAINLY

The submission template anticipates a Next.js and FastAPI web application. Sākṣī is a native Android application, because its core features are hardware features: camera, GPS, magnetometer, accelerometer, on-device inference and durable local storage. A browser cannot deliver an alignment-locked capture with a compass reading attached. We do ship Next.js and Tailwind, for the public landing site, and the knowledge engine is a service the app calls, but it runs on Node rather than Python so that the identical engine file executes on the phone and on the server. Section 5.2 maps each of the template's expected components onto what we actually built and why.

5.1 The layered architecture



Imports go downward only, and everything crosses through a single barrel module rather than reaching into internal files, so the domain layer stays free to be reorganised. Coordinates convert at the seam: the app speaks latitude and longitude, the domain layer speaks lat and lon.

5.2 The stack, component by component

TEMPLATE EXPECTATION, MAPPED TO WHAT WE SHIPPED

COMPONENT	WHAT WE SHIPPED	WHY, IN ONE SENTENCE
Frontend	Expo SDK 57, React Native 0.86, React 19, expo-router with typed routes	Every core feature is hardware-bound, and typed routes turn a broken link into a compile error.
Web presence	Next.js 16, React 19, Tailwind CSS 4	The public landing site, deployed independently of the app.
Local database	SQLite in WAL mode, 8 versioned migrations	A photograph taken at a vantage on a given day cannot be retaken, so the device is the record.
Cloud database	Supabase, PostgreSQL, Row Level Security, 8 migrations	Real relational Postgres, with RLS as the actual protection so a publishable key in the bundle is by design.
AI & logic service	Zero-dependency Node HTTP service importing the same `core/dhamma` engine the tests cover	Chosen over FastAPI so the engine that answers on the server is literally the file that runs on the phone.
Retrieval	Bilara segment-aligned corpus, lexical plus semantic ranking fused by reciprocal rank	Usable recall over a deliberately narrow canon with no index server and no cold start.
Mapping service	MapLibre GL Native, with a WebView implementation and a 3D site plan as fallbacks	Vector tiles, offline styling, no per-view billing, and an open licence appropriate to a public trust.
On-device vision	ONNX Runtime, YOLOv8 crack detector, mAP50 0.82	Damage detection has to work in a garden with no signal, and the photograph should not have to leave the phone.
On-device language	llama.rn with Qwen3 0.6B Q4_K_M, 484 MB, downloaded and checksum-verified rather than bundled	An answer with the radio switched off, without shipping half a gigabyte inside the APK.
Distribution	EAS Build and EAS Update, runtime version pinned to app version	A JavaScript fix ships in seconds; a native change is forced through a new binary instead of crashing an old one.

Because the domain layer is pure TypeScript, 113 tests across 16 files run in about two seconds with no transpiler and no test framework.

5.3 Data flow and content provenance

Content is never typed into components. It lives as reviewable JSON, is compiled into typed TypeScript constants by a generator, and is checked by a validator before it can ship. A change to a coordinate, an evidence tier or a photography policy therefore arrives as a diff with an author attached, which is the audit trail heritage claims require. A startup integrity pass then verifies that every site identifier referenced by a precinct, quest or vantage actually resolves, so that a rename cannot silently disable arrivals across half the Sacred Garden.

5.4 Choices we would defend, and the alternatives

DECISION	WHY WE CHOSE IT	THE STRONGER ALTERNATIVE, HONESTLY
Expo over bare native	Config plugins let us add MapLibre, ONNX Runtime and llama.rn without hand-maintaining an Android project, and EAS produces a signed build from a laptop.	Kotlin would win decisively on camera latency and sensor fusion, which is exactly where our hardest feature lives, at the cost of any iOS path.
React Context over a store	The genuinely global state is six providers of small values; the heavy state lives in SQLite and is queried, not held.	Zustand, and this is the closest call. Selector subscriptions would stop a provider that changes on every GPS tick from re-rendering a subtree.
ONNX over TFLite	It matched the export path straight out of the training notebook, and one runtime serves any future model.	LiteRT is faster on Android through mature GPU and NNAPI delegation, and is the upgrade when latency becomes the complaint.
Rank fusion over embeddings	Usable recall over a narrow canon with no index server, no embedding cost and no cold start.	Real embeddings in pgvector would beat us on paraphrase and on Nepali. It is the first upgrade at ten times the corpus.
Supabase over Firebase	A conservation time-series is relational, and PostGIS is one extension away when coordinates become survey-grade.	Firebase is faster to wire and has the better turnkey offline SDK, and is weaker exactly where this domain is strong.
An append-only merit log	The history is the record; balance, daily cap and cooldown are all derived by reading rows, so they cannot drift out of step with a stored total.	This is also our answer to the blockchain question: merit is non-transferable by construction, so the problem a distributed ledger solves never arises.

5.5 Verification record

Every gate below was executed against the working tree while this document was being written.

GATE	COMMAND	RESULT
Type safety, including typed routes	`tsc --noEmit`	CLEAN
Domain logic, 16 test files	`npm test`	113 / 113 PASS
Seed content and coordinate provenance	`npm run validate`	OK, 5 WARNINGS
Product vocabulary and typography	`npm run vocab`	CLEAN
Knowledge benchmark, 50 questions	`npm run eval:dhamma`	49 / 50

The vocabulary linter is worth a sentence of explanation, because it is unusual. It fails the build if engagement, gambling or game vocabulary reaches a user-facing string. That is how a product principle stays true at the fiftieth commit rather than only in the README, and it caught real drift during the hackathon.

5.6 What is implemented but not yet proven

Unit tests prove domain logic and nothing at all about a compass in a garden. We keep this list explicitly so that nothing in this document overstates the position.

CLAIM	STATUS	WHY IT IS OPEN
Camera, GPS and compass at Lumbini	DEVICE ONLY	Calibration and GPS accuracy under tree cover are field facts, not test facts.
Live sync to a provisioned cloud project	UNCONFIGURED	Client and migrations exist; an end-to-end run against a live project with RLS enabled has not been done.
Full offline behaviour in airplane mode	UNTESTED	Needs the exact build on the exact device with the radio off.
iOS, anything at all	NEVER BUILT	No credentials and no build has ever run. Android only, and we say so rather than implying parity.
Five site coordinates	DOCUMENTARY	Puskarini, Marker Stone, Vihara Remains, Tilaurakot and Ramagrama are documentary, not surveyed. The validator warns on every build.
Provider key handling	KNOWN DEFECT	A public-prefixed key is inlined into the bundle and readable by anyone with the APK. Acceptable only for a scoped, rotatable demo key. Moving it behind the service is the first item on the roadmap.
Device and end-to-end test coverage	ABSENT	The paths most likely to fail on stage are the ones with no automated coverage.

The demo service deserves the same candour: most of its mutable state is in memory, it authenticates nobody, and it should be deleted the day a real API exists. It is a harness that lets the whole offline queue path be exercised on venue wifi with no install step, and it is not infrastructure.

Nothing in the revenue model touches the merit system. Punya has no purchase path, and adding one would break the property it exists to hold.

Business Model & Viability

6.1 Target Audience

SEGMENT	SCALE	WHAT THEY ARE PAYING FOR, IF ANYTHING
International pilgrims	Large, seasonal, high intent	Depth and trustworthiness. This group already pays for guides and is unusually sensitive to whether a claim is sourced.
Domestic and regional travellers	The largest group by headcount	Little or nothing. This tier must stay free and must work on a mid-range Android phone with poor connectivity.
Heritage authorities	LDT, municipalities, Department of Archaeology	A continuous condition record and a triage dashboard they would otherwise have to staff.
Institutions	Universities, conservation NGOs, monastic bodies	Access to the longitudinal dataset, and the ability to define their own vantages.

6.2 Revenue Streams

FREEMIUM TIERS ON TĪRTHA

The free tier is not a demo. It carries the full map, arrival detection, basic and medium wisdom, the complete witness workflow, and unrestricted use of the knowledge surface. This is a deliberate constraint: the conservation value of the product depends on volume of observations, so anything that reduces the number of people capturing is self-defeating. The paid tier sells depth, not access: curated multi-day routes across the Greater Lumbini Circuit, offline map and narration bundles, the high wisdom tier with its epigraphic and source detail, and multilingual audio guides as they are produced. A guided-circuit pass sold seasonally to international visitors is the realistic first revenue line.

B2G LICENSING ON SĀKSĪ

This is the durable business. The civic dashboard is licensed per municipality or per heritage authority: the condition register, severity and corroboration triage, the observation time-series at each vantage, exports in CSV and GeoJSON, and CRM-shaped output for whatever case system the authority already runs. The pricing argument is straightforward, because the counterfactual is a commissioned survey. A licence that costs a fraction of one specialist survey and produces a continuous record instead of a single snapshot is an easy comparison to make, and it is made against a budget line that already exists.

ETHICAL B2B SPONSORSHIP

Sponsorship exists, and it is fenced. Named restoration and conservation needs can be sponsored by institutions, monastic bodies and businesses, with acknowledgement rendered as a line in a register rather than as advertising placed between a visitor and a monument. Three rules hold: no interstitials, no sponsored content inside the knowledge surface, and no sponsor influence on what the condition data says. The directed dāna mechanism already models allocation against named needs in the codebase, without moving real money, so the shape of this is built and the payments integration is deferred rather than the concept being speculative.

> WHAT WE WILL NOT SELL
>
> Visitor location traces, merit points or any in-app currency, ranking positions on the leaderboard, and placement inside a Dhamma answer. Each of these would generate revenue and each would destroy the property that makes the corresponding feature worth having. The condition dataset is licensed to authorities and researchers, and the underlying observation record belongs to the heritage authority, not to us.

6.3 Impact Metric

The headline metric is **monitored vantage coverage**: the number of established viewpoints receiving at least one aligned capture per month, and the median interval between captures at each. That single number is the difference between an episodic survey regime and a continuous one, and it is the number a heritage authority can act on. Secondary metrics follow from it: time from a condition first appearing in imagery to a report being filed, corroboration rate, and the proportion of reports that reach a resolution.

The benefit to the local ecosystem in Nepal is meant to be structural rather than charitable. The dataset is generated in Nepal, held for the authority responsible for the site, and licensed on terms that keep it there. Content production, that is narration, translation into Nepali and regional languages, and transcription, is work that is credited and compensable, and the merit system already recognises contribution as a distinct earning category. Local guides are complemented rather than displaced: the app handles the material a guide should not have to repeat forty times a day, and directs paid demand toward the depth only a person can provide. Above all, a heritage authority that can see a crack widening across six months of comparable photographs can intervene while the intervention is still cheap, which is the whole economic argument for preventive conservation.

7

CHALLENGES

Each of these is a bug we shipped and then fixed, not a hypothetical. The fixes are all in the repository history.

Challenges Faced During the Hackathon

7.1 Technical Hurdles

THE ENVIRONMENT VARIABLE THAT WAS NEVER READ

The knowledge engine read its provider key from an unprefixed environment variable. Expo only inlines variables carrying the public prefix into the JavaScript bundle, so on every physical device the key was undefined and the model provider was never called at all. The system did not fail: it fell through to the deterministic cited fallback and returned a plausible, correctly cited answer every time, which is precisely why nobody noticed. A graceful degradation path can hide a total outage of the thing being degraded from. We now assert the provider path explicitly rather than inferring it from a good-looking answer, and the provider note is part of the response we check.

A DAMAGE DETECTOR THAT INVENTED ITS OWN FINDINGS

The first version of the on-device damage detector hashed the image *filename* into deterministic bounding boxes, confidence scores and a "surface integrity" percentage. It looked convincing in a demo and it was fabrication. We replaced it with a real YOLOv8 detector exported to ONNX. The runtime returns a raw tensor, so the letterboxing, decoding, non-maximum suppression and mapping back to original photograph pixels are all explicit pure functions with their own tests rather than hidden inside a helper. The rule we adopted afterwards is that every exit path from the detector must carry a reason string that reaches the screen, because honest-by-silence is indistinguishable from a feature that was never built.

THE RENAME THAT BROKE ARRIVALS WITHOUT BREAKING ANYTHING

A parallel branch migrated the site list onto generated seed data and renamed identifiers on the way: `ashoka-pillar` became `ashokan-pillar`, `puskarini-pond` became `puskarini`, and two sites were dropped. Nothing in the type system noticed. The site lookup returned undefined, every caller skipped it, and arrivals silently stopped firing across three quarters of the Sacred Garden while continuing to look like they worked. Git was perfectly satisfied. We now run a referential integrity pass over the content at startup that fails loudly when a precinct, quest or vantage references a site that does not resolve.

HALF A GIGABYTE, REHASHED ON EVERY QUESTION

The offline language model is a 484 MB file downloaded into app-private storage rather than bundled into the APK. The first implementation verified its checksum to answer the question "is this file the model", which meant hashing 484 MB before every answer, and a download that failed at 90 percent restarted from zero. The fix was a resumable download plus a marker file written once after verification succeeds, so the question can be answered without re-reading the file.

OPTIONAL NATIVE MODULES AND THE BUNDLER

ONNX Runtime, llama.rn, MapLibre, speech and notifications are native modules that exist only in a build that installed them. A static import of an absent package makes the bundler fail to build the entire application: not the feature, the application, including for a teammate who simply had not rebuilt yet, and including for an over-the-air update landing on an older binary. Every optional native module is now loaded through a guarded runtime require that resolves to null, so one feature reports an honest unsupported state and every other surface keeps running.

REAL-TIME MAPPING

Two mapping problems took real time. The first was the native and web split: MapLibre's native module has no web equivalent, so a WebView implementation and a hand-drawn 3D site plan sit behind the same component interface. The second was geofence chatter. A naive radius test fires enter and exit repeatedly when a visitor stands near a boundary with normal GPS jitter, producing a stream of notifications on consecrated ground. Hysteresis solved it: enter at the radius, exit only at 1.15 times the radius, with a dwell timer for the events that should require actually staying.

7.2 Validation: ensuring the authenticity of Dhamma text processing

Authenticity here is not a matter of good intentions, so we made it measurable.

1. **Segment-aligned canonical source.** The corpus is built from Bilara segment-aligned texts, which means every passage carries a canonical identifier such as `dn16:6.7`. This is the load-bearing decision: a reader can open that exact segment and check us. A corpus of loose paragraphs would make every downstream guarantee unverifiable.
2. **A deliberately narrow canon.** Admitting more material makes the system able to answer more questions and much likelier to blend traditions without saying so. We anchored on the Dhammacakkappavattana Sutta, the Mahāparinibbāna Sutta and the Kālāma Sutta with a small set of related discourses, and we accept refusals as the cost.
3. **Validation after generation, not only before.** Every claim in a drafted answer is checked against the segment identifiers actually retrieved. An unvalidatable answer becomes a refusal rather than a hedge, which is what makes the refusal figures meaningful.
4. **An adversarial suite, written to break our own system.** Fifty questions in five categories: answerable, adjacent to the corpus edge, out of scope, adversarial, and Nepali. Adjacent is the hardest category and the one we

still lose a point on, because it is where over-answering is most tempting. Adversarial covers prompt injection and requests for the system to speak as the Buddha in the first person, and it passes at 100 percent, which was a mandatory gate rather than a target.

5. **A Nepali distress override that bypasses everything.** A Nepali self-harm or distress query does not go anywhere near retrieval. It returns helpline information. Treating that as a retrieval problem would have been a serious failure, and it is the one path in the system with no fallback, no degradation and no clever behaviour.
6. **Language parity checks.** Nepali is the default answer language, and hosted models are markedly weaker in Nepali than English. Citations and source evidence are attached identically in both languages, and the language toggle re-requests the same question rather than machine-translating an English answer, so a Nepali reader is never given a translation of a claim that was only ever validated in English.

Phase 1 is deliberately unglamorous. Nothing else is worth doing until the coordinates are survey-grade and the key is off the client.

Future Roadmap

Phase 1 — Immediate: a pilot in the Greater Lumbini Circuit

The goal of this phase is not new features. It is to make the existing path trustworthy enough that a heritage authority can rely on what comes out of it.

1. **Move the model provider key behind the service.** One environment variable and one request path. This closes the only genuine security defect in the current build and it is the first thing we do.
2. **Replace the demo service with a real API.** Authentication, durable storage, rate limiting and monitoring, then delete the harness rather than letting it drift into production by accident.
3. **Establish survey-grade vantages with the Lumbini Development Trust.** This is the input only they can provide, and it is what converts the app from a prototype into an instrument. Five site coordinates are currently documentary rather than surveyed, and the vantage points are not yet established viewpoints. Until both are fixed, the app should keep saying so on screen.
4. **Field-test on the actual ground.** GPS accuracy under tree cover in the Sacred Garden, compass calibration, capture in bright daylight, and the complete flow in airplane mode.
5. **Add device-level test coverage** for the capture, permission and offline flows, since that is where a live failure will come from.
6. **Retrain the crack detector** on Lumbini brick and sandstone specifically. A single-class model at 0.82 mAP50 demonstrates the pipeline; it is not yet a conservation instrument.
7. **Run a bounded pilot**: a defined set of vantages, a small cohort of repeat observers, and a monthly report to the Trust, so that the impact metric has a real baseline.

Phase 2 — Mid-term: the civic dashboard, beyond Lumbini

- **Municipal onboarding.** A heritage officer defines their own sites, vantages and geofences without an engineer, which is the precondition for the product working anywhere we are not personally standing.
- **Triage and workflow.** Reports ranked by severity, corroboration count and rate of change; assignment, resolution and a resolved-condition record so that the register shows repair as well as damage.
- **Change detection across the series.** Automatic flagging when consecutive captures at the same vantage differ beyond a threshold, which turns the archive from a lookup into an alert.
- **Data custody and export.** Formal agreements placing the observation record with the heritage authority, plus CSV, GeoJSON and CRM-shaped

export into whatever case system already exists.

- **Migrate retrieval to real embeddings** in pgvector as the corpus outgrows rank fusion, and move the hosted model to a provider materially stronger in Nepali. The latter is a single-file change through an existing seam.
- **iOS.** Not yet begun, and honestly out of reach until Phase 1 is complete.

Phase 3 — Long-term: the Kathmandu Valley Circuit

- **Scale to the Valley.** Swayambhunath, Boudhanath, Pashupatinath and the three Durbar Squares, which are denser, more contested and under far heavier visitor pressure than Lumbini. Post-2015 reconstruction makes a continuous photographic record particularly valuable there.
- **Multilingual audio guides.** Nepali, English, Hindi, Chinese, Japanese, Korean, Thai, Sinhala and Burmese, matching the actual pilgrimage traffic. Recorded by people rather than synthesised where the material is devotional, and credited as contribution work.
- **Open the dataset for research.** A longitudinal, positionally consistent photographic archive of South Asian heritage monuments is a genuinely novel research asset. Licensed openly for academic use, held in Nepal.
- **A vantage standard others can adopt.** The alignment scoring, the metadata schema and the honesty rules generalise well beyond Buddhist heritage, and are more valuable as a published specification than as our private implementation.

Four members. LumbiniX Committee, Nepal.

The Team

Aaditya Sapkota

AI BACKEND ENGINEER

Owns the knowledge engine end to end: the Bilara corpus loader, hybrid retrieval and reciprocal rank fusion, the grounding gate, the citation validator, the deterministic fallback, and the 50-question adversarial benchmark. Also responsible for the on-device inference path, covering the ONNX vision runtime and the quantised offline language model.

Binayak Gautam

3D RENDERING & FRONTEND ENGINEER

Owns the visual surface: the MapLibre implementation with its native, web and 3D site-plan variants, monument massing and site plans, the Then and Now comparison and scrubber, the capture and alignment reticle, and the theme and component system that keeps colour and type out of individual screens.

Nabin Poudel

RESEARCH & DOCUMENTATION

Owns content authenticity: site research, evidence tiers and coordinate provenance, historical plate licensing, the canonical corpus selection and its narrowness, Nepali language review, and this document. Also maintains the seed validator warnings that keep documentary coordinates from being presented as surveyed.

Siddanta Sodari

FULL STACK DEVELOPER

Owns the application architecture and the platform: the layered structure and its purity rule, SQLite schema and migrations, the offline queue and cloud sync, permissions and sensor boundaries, the alignment scoring and merit systems, the demo service, the landing site, and the EAS build and update pipeline.

Repository & Live Demo Links

GITHUB REPOSITORY	https://github.com/LumbiniX-Committee/Everest Full source: the Expo application, the pure domain layer, the knowledge engine, seed content, database migrations, tooling and the landing site.
LIVE MVP / PROTOTYPE	application-a925ec89-66ca-48ae-a9c8-744067a46082.apk Installable Android build produced by EAS. Camera, location and motion sensors require a physical device; the map, knowledge surface and register work in an emulator.
DEMO VIDEO	https://drive.google.com/drive/folders/1by7PGKVBpTmufWhx-YoBqBDkdZ02BofS Walkthrough of the three surfaces: arrival and tiered wisdom in Tīrtha, the alignment and capture loop in Sākṣī, and a cited answer plus a refusal in Dhamma.

CLOSING NOTE

Most of the architecture in this document exists to hold one property: the application never claims more than it can support. A match by eye is not a sensor lock. A missing reading is not a zero. An uncited claim is not an answer. A detected crack is not a report until a person confirms it. A documentary coordinate is not a survey. Those are five separate mechanisms in five separate subsystems, and they are all the same decision, taken from the same instruction: do not accept a thing because it is said confidently. Test it, and see for yourself.

साक्षी

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